



The Ephemeris



December 2016

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Dec - Feb 2017 Events

Board & General Meetings

Saturday 12/3, 1/14, 2/11, 3/11

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 12/04, 1/8, 2/5, 3/5

Solar Observing (locations differ)

Sunday 12/04, 1/8, 2/5, 3/5

Intro to the Night Sky Class

Houge Park 1Q In-Town Star Party

Friday 12/02, 1/6, 2/3, 3/3

Astronomy 101 Class

Houge Park 3Q In-Town Star Party

Friday 12/16, 1/20, 2/17, 3/17

RCDO Starry Nights Star Party

Saturday 12/17, 1/21, 2/18, 3/18

Imaging SIG Mtg

Tuesday 12/20, 1/17, 2/21, 3/21

Astro Imaging Workshops

(at Houge Park)

Saturday 3/25, 6/24

Coder's Meeting

Sunday 12/18, 2/19, 3/19

Quick STArT (by appointment)

Friday 12/9, 2/10, 4/7

Binocular Star Gazing

Program will resume Saturday 5/20

Unless noted above, please refer to the SJAA Web page for specific event times, locations and possible cancellation due to weather.

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Cover Photo: Helix Nebula
NGC 7293

By SJAA Member: Bob Thompson



Photo Credits: Ed Wong



Happy 80th Jim!!

SJAA members surprised Jim Van Nuland with an 80th Birthday party at the October 15th general meeting. The event was well attended, everyone had a blast and Jim appeared truly surprised.

Jim has been a member of SJAA since well, forever.

His participation in the SJAA has spanned over 30 years and his many contributions include running the School Star Party Program for the duration, Club Secretary for 20 years or more, he created a program for recording the annual auction results, ran the auction, developed the club's annual calendar for many years and the list goes on.



2016 Ends with Three Supermoons

The term supermoon has entered popular consciousness in recent years. Originally a term from modern astrology for a new or full moon that occurs when the moon is within 90% of its closest approach to Earth in a given orbit, supermoon now refers more broadly to a full moon that is closer to Earth than average. But why is the moon closer to Earth at some times but not others?

Since the moon's orbit is elliptical, one side (perigee) is about 30,000 miles (50,000 km) closer to Earth than the other (apogee). The word syzygy, in addition to being useful in word games, is the scientific name for when the Earth, sun, and moon line up as the moon orbits Earth. When perigee-syzygy of the Earth-moon-sun system occurs and the moon is on the opposite side of the Earth from the sun, we get a perigee moon or more commonly, a supermoon!

This coincidence happens three times in 2016. On October 16 and December 14, the moon becomes full on the same day as perigee. On November 14, it becomes full within about two hours of perigee—arguably making it an extra-super moon.

The full moon of November 14 is not only the closest full moon of 2016 but also the closest full moon to date in the 21st century. The full moon won't come this close to Earth again until November 25, 2034.

The supermoon of December 14 is remarkable for a different reason: it's going to wipe out the view of the Geminid meteor shower. Bright moonlight will reduce the visibility of faint meteors five to ten fold, transforming the usually fantastic Geminids into an astronomical footnote. Sky watchers will be lucky to see a dozen Geminids per hour when the shower peaks. Oh well, at least the moon will be remarkable.

How remarkable? A supermoon, or perigee full moon can be as much as 14% bigger and 30% brighter than an apogee full moon. However it's not always easy to tell the difference. A 30% difference in brightness can easily be masked by clouds or the competing glare of urban lights. Also, there are no rulers floating in the sky to measure lunar diameters. Hanging high overhead with no reference points to provide a sense of scale, one full moon looks much like any other.

Low-hanging moons, on the other hand, can create what's called a "moon illusion." When the moon is near the horizon it can look unnaturally large when viewed through trees, buildings, or other foreground objects. The effect is an optical illusion, but that fact doesn't take away from the experience.

A supermoon is undeniably beautiful. And we can multiply that beauty by three as 2016 comes to a close.

December 14: Mark your calendar and enjoy the super moonlight.

Credit: Nasa.gov

Planet Orbiting Nearby Star Could be Habitable

Key Points— A recently discovered planet orbiting our nearest neighboring star, Proxima Centauri, could be habitable.

A French-American team of researchers created simulations of the composition of the planet, Proxima b. They found that it could have a rocky surface with water, or even a surface completely covered by a single ocean.



This artist's impression shows a view of the surface of the planet Proxima b orbiting the red dwarf star Proxima Centauri, the closest star to the solar system. The double star Alpha Centauri AB also appears in the image. Proxima b is a little more massive than the Earth and orbits in the habitable zone around Proxima Centauri, where the temperature is suitable for liquid water to exist on its surface.

Credit: ESO/M. Kornmesser

A rocky extrasolar planet with a mass similar to Earth's was re-

cently detected around Proxima Centauri, the nearest star to our sun. This planet, called Proxima b, is in an orbit that would allow it to have liquid water on its surface, thus raising the question of its habitability. In a study to be published in *The Astrophysical Journal Letters*, an international team led by researchers at the Marseille Astrophysics Laboratory (CNRS/Aix-Marseille Université) has determined the planet's dimensions and properties of its surface, which actually favor its habitability.

The team says Proxima b could be an "ocean planet," with an ocean covering its entire surface, the water perhaps similar to that of subsurface oceans detected inside icy moons around Jupiter and Saturn. The researchers also show that Proxima b's composition might resemble Mercury's, with a metal core making up two-thirds of the mass of the planet. These results provide the basis for future studies to determine the habitability of Proxima b. Credit: Nasa.gov



Hubble Nets a Subtle Swarm

This Hubble image shows NGC 4789A, a dwarf irregular galaxy in the constellation of Coma Berenices. It certainly lives up to its name — the stars that call this galaxy home are smeared out across the sky in an apparently disorderly and irregular jumble, giving NGC 4789A a far more subtle and abstract appearance than its glitzy spiral and elliptical cousins.

These stars may look as if they have been randomly sprinkled on the sky, but they are all held together by gravity. The colors in this image have been deliberately exaggerated to emphasize the mix of blue and red stars. The blue stars are bright, hot and massive stars that have formed relatively recently, whereas the red stars are much older. The presence of both tells us that stars have been forming in this galaxy throughout its history.

At a distance of just over 14 million light-years away NGC 4789A is relatively close to us, allowing us to see many of the individual stars within its bounds. This image also reveals numerous other galaxies, far more distant, that appear as fuzzy shapes spread across the image.

Image Credit: ESA/Hubble & NASA, Acknowledgements: Judy Schmidt (Geckzilla)

Text Credit: European Space Agency

OSA Coyote Valley Harvest Festival September 10, 2016

From Bill O'Neil



Wolf's investment in colorful UV Beads was a huge crowd pleaser for families with young kids, making bracelets under the canopy. See Susan's Photos attached.

Rob Jaworski, Wolf, Marion & Paul also setup a Solar Party at the Cambrian Festival in San Jose on Sunday Aug. 28. This is our 6th year there and is a show of appreciation to the City of San Jose for allowing SJAA to use Houge Park.



We had a very successful OSA Coyote Valley Harvest Festival the weekend of September 10th.

Thanks to Wolf, Marion & Paul, Gary, Terry, David B., Kevin, and new member Matt Welsh for bringing solar telescopes and/or helping with the over 1200 visitors that day.

Glenn & Bill setup the 19" Bright monitor, plus MallinCam Sky-Raider camera and the new Double Stack on the club's LS100THA. FYI, I posted my "Fun Facts About Our Sun" on the Solar.net google drive.

Everything worked great (although tuning two etalons is a work in progress. Glenn's two 35Ah batteries powered the computer, and TV monitor for over 5 hours. Which means we can use this camera & TV setup for the Boy Scout-O-Rama in May, and the Cupertino Earth Day on April 22, 2017.



Bill O'Neil
SJAA Director
Solar Observing, Education & Outreach

Photo Credits: Susan O'Neil

SJAA Member Observation Posts

Glacier National Park September 3, 2016

From Sukhada Palav

It was a wonderful watching the sky from Glacier NP! They were hosting a nightly astronomy program at St. Mary Visitor center which was 10 minutes drive away from our hotel. So we decided to take the opportunity of the day we got there. The weather forecast for next 2 nights was cloudy and rainy.

We got there a little late, "What's up" talk had already started. After the talk, I went and introduced myself to 2 gentlemen hosting the event - Steve and Ray. They are Bill's friends so had to convey his regards too! ;-) @Bill - they say hi back! :-)

There were 2 scopes, one 10" Dobsonian and one refractor set up. We looked at M13, Wild duck, ET cluster, double cluster and few more objects. There were easily 50 visitors that night. I carried my binoculars so I enjoyed views of double cluster and Andromeda galaxy among other things while my friends were completely focused on trying to capture the Milky Way in their cameras! Can't blame them...Milky Way from there looked better than anything I have seen from around here...even Pinnacles. I still wouldn't call the visitor center parking lot a perfect site for two reasons - 1) there was some glow coming in from the east side. 2) Ray told us bears frequent this parking lot! ;-) But you know, that's true for any place in this park...

Also, Ray told us that there was a chance of seeing Auroras that night, between 11pm and 1am! Bill had already mentioned to me that the chances are better on the east side of the park. So we waited...and waited... Ray kept saying he can see some light activity on the horizon but none of us could see it. He told us his friend in NY is messaging him that the solar wind is dying down. So it didn't turn out to be the spectacular show we were hoping for.

Finally, just after midnight we decided to leave and made it back to our hotel at 12:30am. We had been up since 3:30am that day so it was LONG day but I am glad we decided to stretch ourselves and attended the program!

Review of Vixen's SG 2.1x42 wide-field binoculars

From Marilyn Perry

Brian Ventrudo of Cosmic Pursuits posted a good review of a new kind of star gazing gadget, Vixen's SG 2.1x42 wide-field binoculars. They are light enough for me to lift and small enough to fit in my house unlike some star gazing apparatus I would like to have, so I bought a pair.

They look weird, like extremely thick glasses or like the goggles worn by cantina owner Maz Kanata in Star Wars, The Force Awakens, but they work well for me and I am enjoying them. The Cosmic Pursuits article praises their ability in dark skies, but I have not yet had that opportunity.

I am getting my money's worth by using them to help in star hopping in my light-polluted backyard by helping me identify stars. For example, with unaided eye, I can clearly see only one star in Cepheus, mag 2.5 Alderamin. With no context to put it in, I was not sure what star that was. But with my Vixen 2.1x42 binos and their super-wide field of view, I can put the entire stick figure of Cepheus in the field at once and see all of the stars that make up the stick figure. The Cosmic Pursuit article says that size of the FOV depends on where you put their eyes. Since I can see entire Cepheus stick figure, I am guessing at least 21 degrees.

It is a little disconcerting that stars aren't in sharp focus all the way to the edge like with my Televue Delite eyepieces, but I put up with that inconvenience for the pleasure of seeing the entire constellation. At 2.1x the magnification is so small that I don't even have the sense that things are being magnified; it just seems like I have "super-vision" and now I can see many more stars. I eagerly look forward to taking them out in dark skies to see how the Milky Way appears through them.

Here is link to info about these binoculars:

<http://cosmicpursuits.com/1191/vixen-sg-2-1x42-binoculars/>

SJAA Member Observation Posts

Observing Close Conjunction of Moon and Aldebaran + Occultation of 89 Tau

From Marilyn Perry



Close Conjunction of Aldebaran and the moon
October 18, 2016, 22:30 PDT
Canon T3i, Stellarvue 102mm refractor, ISO 800, 1/320 sec

In preparation for the evening of October 18, 2016 when much of the US could view the occultation of Aldebaran by the moon but Northern Californians could not, I consulted Sky Safari to see if any of the other bright stars in the Hyades would be occulted and found that mag 5.8 89 Tau would be occulted from about 23:03 PDT until about 23:22 PDT.

I used my Canon T3i and StellarVue 102mm refractor to photograph the close conjunction of the moon and Aldebaran. With ISO 800, shutter speed 1/320 second, I was able to capture mag.1.0 Aldebaran as well as craters near the terminator plus maria on the surface, but no other stars.

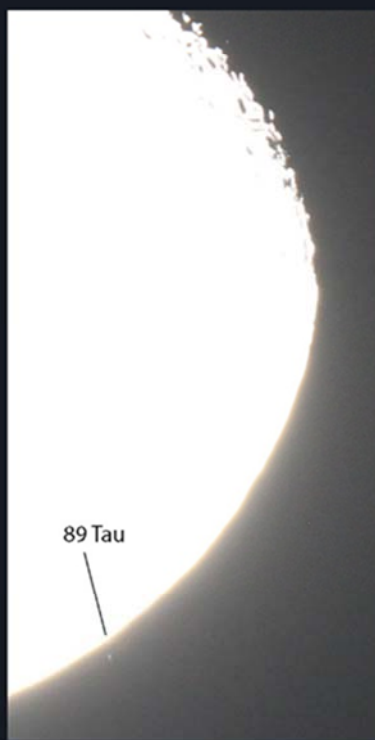
Comparing images taken 1 minute apart, I think Aldebaran was closest at 22:30 PDT. I found that if I slowed shutter speed to 1/13 second, I could also capture 89 Tau. Sigma 1 and 2 Tau, both mag 4.7, and Aldebaran showed up in live view. The

surface of the moon was totally blown out, although some craters right at the terminator were visible.

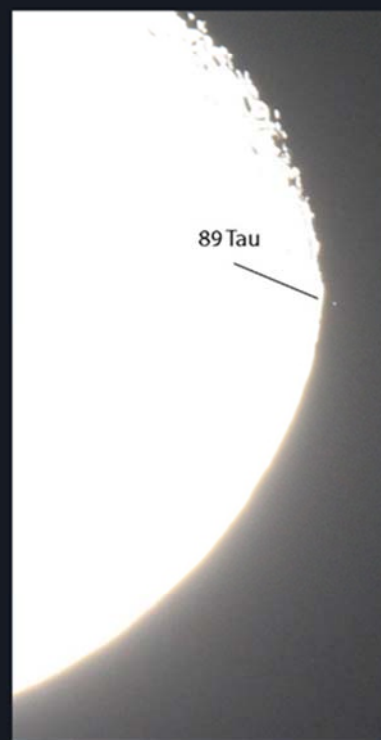
I thought it would be more fun to visually observe 89 Tau disappear behind the moon, so I switched to visual. It didn't matter whether I used high or low power or how much of the moon was visible in the eyepiece, with 89 Tau so close to the glare of the moon I could not see it. However I could see the Sigmas, so I knew I was in the right spot.

I didn't want to miss the occultation, so I switched back to imaging. I could not see 89 Tau in live view but it was visible when I zoomed in on the images. At 23:01 PDT 89 Tau was nicely separated from the glow of the moon; at 23:02, overlapped the glow; at 23:03, no trace; at 11:22 still no trace; at 11:23 nicely separated from the glow.

Afterwards I discovered that the RASC handbook lists occultations of bright stars by the moon, but it did not list 89 Tau for 2016. It does include it for 2017, so 89 Tau is bright enough to be included. I am glad I consulted Sky Safari, and did not rely on RASC because 89 Tau was clearly occulted from my backyard. I guess location makes a big difference, and my backyard is not the same as the RASC "Northern California" location.



89 Tau 3 minutes before ingress
Oct 18, 2016, 23:00 PDT



89 Tau 2 minutes after egress from behind
Oct 18, 2016, 23:24 PDT

Canon T3i, StellarVue 102mm refractor, ISO 800, 1/13 sec exposures

SJAA Member Observation Posts

2016 Oregon Star Party (OSP)

From Gary Chock

I attended the 2016 Oregon Star Party (8/2 to 8/7), arriving early on Sunday 7/31 and staying for a week until Sunday 8/7. The event is held in Central Oregon, east of Prineville, in the middle of the Ochoco National Forest. It is about a 600 mile drive from San Jose to Prineville. From Prineville it takes about 70 minutes to get to the site. The last 5 miles was gravel road but the route is all vehicle access, including cars, RVs, trailers.

<http://oregonstarparty.org/>

<https://www.google.com/maps/@44.2988,-120.1416,1008m/data=!3m1!1e3>

The site is along an east-west road (800 Road) with a spur loop road to the north (802 Road and Loop), all graveled. The terrain slopes away slightly from those roads with good horizons broken up only by a sparse tree line.

The site is on the grey-black boarder of light pollution maps and I only noticed some light dome from Prineville low on the western horizon. The site is at 5,000 feet.

There were 400 pre-registered attendees with 190 on-site registrations. I setup with the Astro-photography folks (west end of 800 Road, south side) because of the great south horizon. We were initially spread out, but as OSP started, more and more folks showed up and filled all the possible gaps between. So it got pretty tight - you get to know your neighbors!

OSP offers great facilities (which mitigate the impact of so many folks). Portable toilets are deployed all around. They have hand washing stations and are cleaned out by the vendor every day (just like new). You can pay for a hot shower (great luxury). There's an espresso / smoothie truck and a food truck. The food truck offers a la carte breakfast and lunch. For dinners, you can pre-pay at pre-registration or try for available a la carte offerings.

The ground is littered with rocks (volcanic) and dusty dirt, but OSP has used the site for so many years that many clear patches can be found for tents and telescopes. Note

that tent stakes do not go deeper than 6 inches, so the tent and shelter tie-downs of choice are 5 gallon buckets filled with those rocks or large water containers. Beware of after-

noon dust-devils that form as breezes meet at the top of this plateau. I witnessed two shelters being lifted 20 to 30 feet off the ground and travel the same distance even with multiple 2.5 gallon water jugs attached to their legs. A large SCT was pushed over by a dust-devil as well.

OSP offers several talks during the main 4 days, this year focusing on the 2017 Total Solar Eclipse and OSU's contribution to the LIGO Gravitational Wave discovery.

There was a Telescope Walkabout featuring some interesting projects - the folks in Northwest seem to be really focused on amateur telescope making (when it is dark and rainy in the winter, what else is there to do?). OSP offer kids activities and telescope mentoring. Plus there's a big raffle at the end.



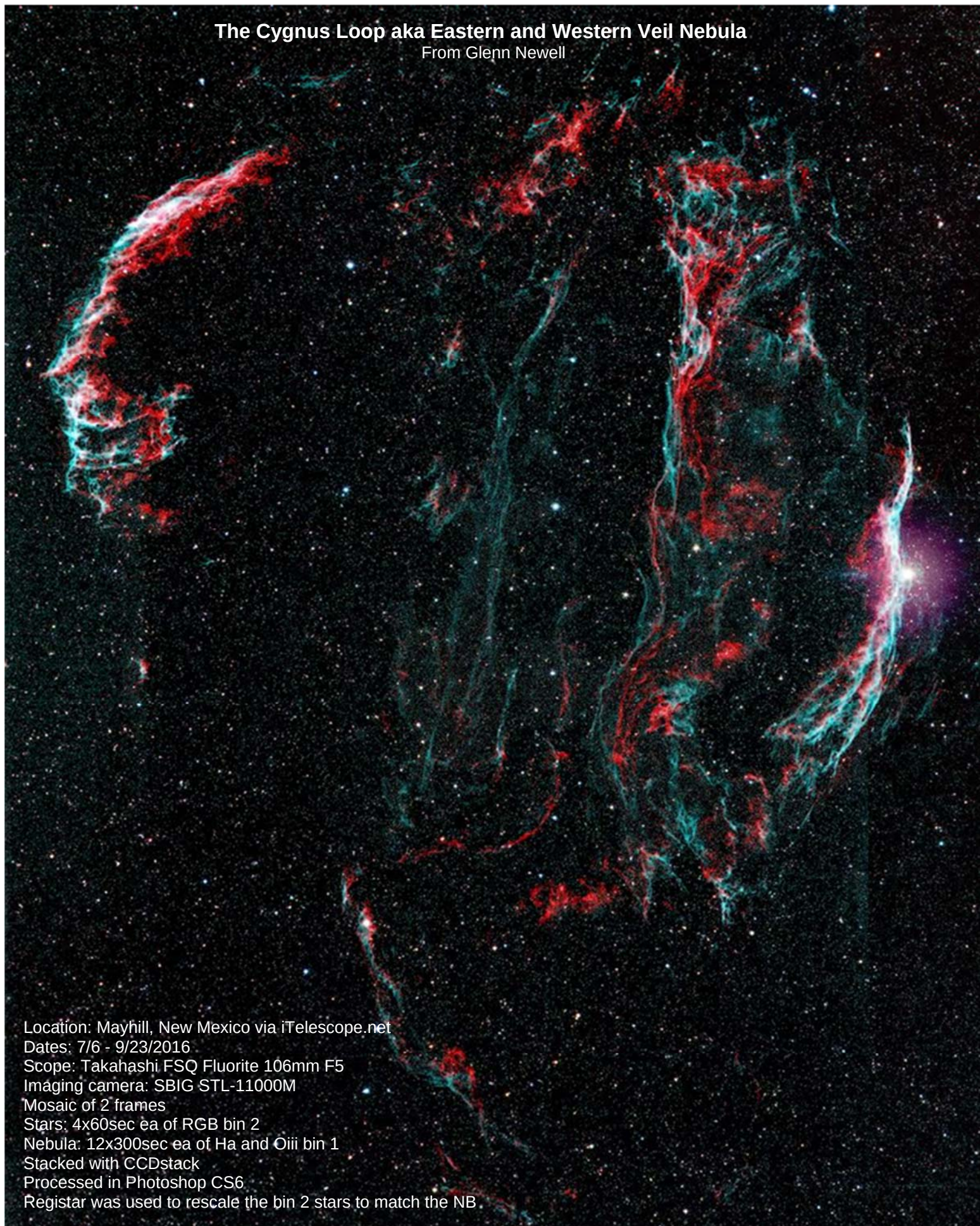
Out of 7 nights, one was completely cloudy with dry lightning and thunder. Another two nights had a just a passing patchy clouds. The remaining were clear. Nights were calm and mild in the 40's. Days only got as high as the upper 80's. Seeing was usually good to very good. Transparency also ranged from good to very good.

My goal for attending OSP 2016 was to scout out central Oregon for the 2017 Total Solar Eclipse on the morning of 8/21/2017. I ventured along Highway 26 to check out John Day Fossil Beds National Monument and Painted Hills. Those looked promising, but the two lane highway and limited infrastructure will make them problematic. For now, I am inclined to attend OSP 2017 - being held from 8/17 to 8/22 in 2017. The OSP site is off the center-line, so instead of about 2:00 minutes, totality would be about 1:30 minutes. I figure I would be among a lot of like-minded folks, by then settled-in instead of fighting traffic and looking for parking. At about two minutes totality, this TSE is short and the experts recommended that first-timers focus on just enjoying it and the full sensory experience without being distracted by equipment.

Photo credit: Gary Chock

The Cygnus Loop aka Eastern and Western Veil Nebula

From Glenn Newell



Location: Mayhill, New Mexico via iTelescope.net

Dates: 7/6 - 9/23/2016

Scope: Takahashi FSQ Fluorite 106mm F5

Imaging camera: SBIG STL-11000M

Mosaic of 2 frames

Stars: 4x60sec ea of RGB bin 2

Nebula: 12x300sec ea of Ha and Oiii bin 1

Stacked with CCDstack

Processed in Photoshop CS6

Registar was used to rescale the bin 2 stars to match the NB.

Location: Manzanita Observatory near Yosemite, California



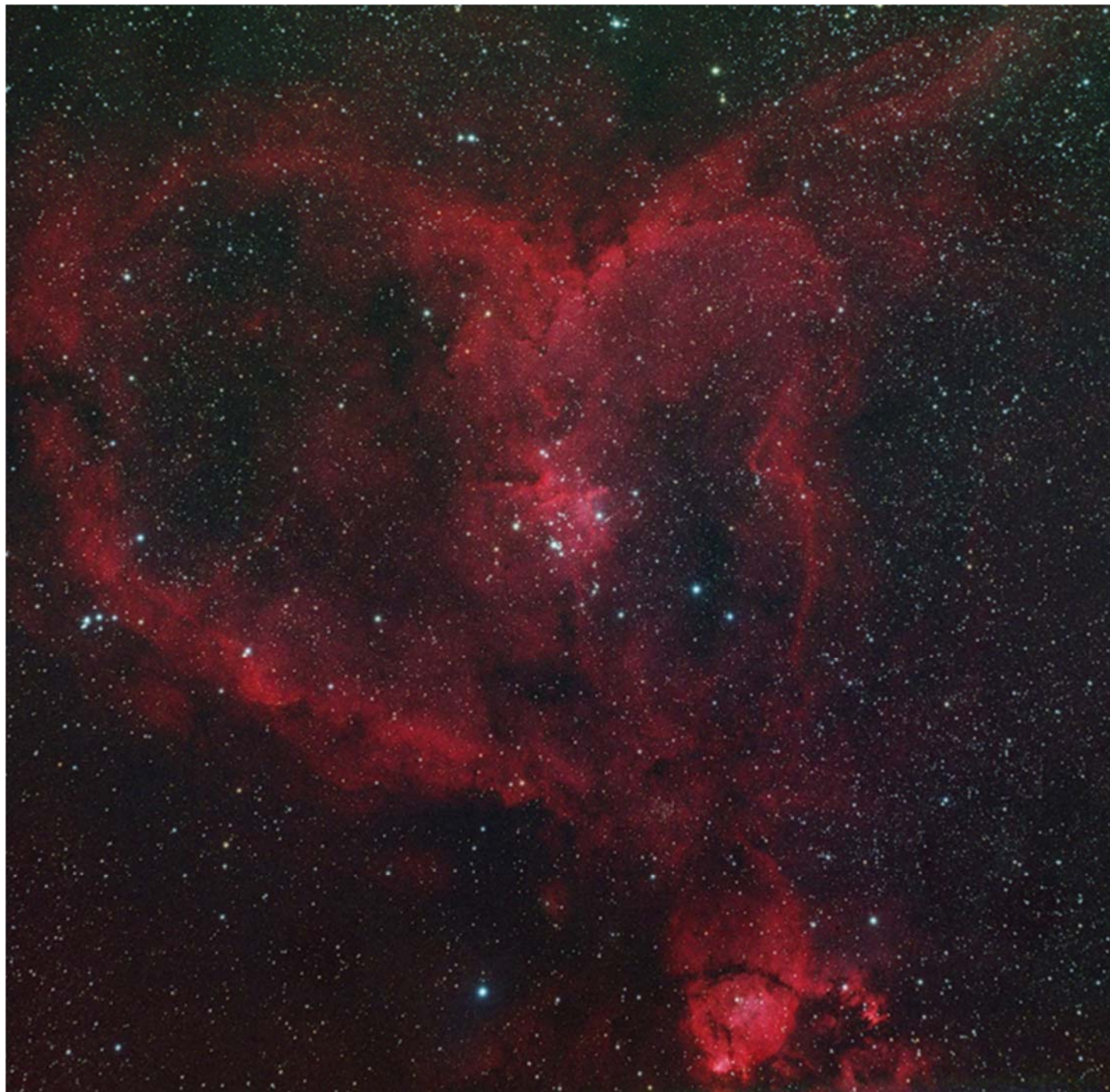
Subject: **M33 and M51**

Camera: Canon DLSR 60Da,
Telescope: Celestron RASA 11" F 2.2
Guiding: PHD with 50 mm Orion Auto guild
Mount: Orion EQ8 mount

From: **PJ Mahany**

Frames:
M33 - 66 @ 180 seconds, Master Dark, Superbias, & Flats.
M51 - 124 @ 180 seconds, Master Dark, Superbias, & Flats.
Software: SkyTools Pro, BYEOS, StarTools.
Processing: PixInsight1.8





Heart nebula IC 1805, Mel 15

From Bob Thomson

Location: Gilroy backyard

Telescope: Takahashi TOA 130 with TOA-67 field flattener @ F 7.7

Mount: Astro Physics Mach One GTO

Camera: SBIG STX-16803 self guided

Filters: Baader

Frames: 4 Hours Ha, 30 min Sub exposures

1 hour RGB, 5 min exposures

FOV just over 2 degrees



Kid Spot Jokes:

- ♦ **Where do astronauts grab a drink?**
At the spacebar!
- ♦ **How do you know when the moon is going broke?**
When it's down to the last quarter.
- ♦ **Where would an astronaut park his space ship?**
At a parking meteor!

Kid Spot Quiz:

1. **What is the correct alignment during a solar eclipse; Sun, Moon, Earth or Sun, Earth, Moon or Moon, Sun, Earth?**
2. **What is the visible part of the Sun called?**
3. **What term describes the alignment of three celestial bodies?**
4. **Who invented the telescope?**

Did You Know?

Source: Astronomy Magazine March 2015 issue "500 Coolest Things"

Saturn's icy moon Iapetus is susceptible to landslides up to 50 miles long.

American Astronomer Edwin Hubble invented the tuning fork diagram in 1936 to classify galaxies by their appearance.

The Soviet Venera 7 was the first spacecraft to land successfully on another planet - Venus - December 15, 1970.

Each of the two opposing beams of the Large Hadron Collider typically contains 2,808 bunches with some 100 billion protons in each bunch.

The largest constellation in the sky, Hydra the water Snake, is 19 times bigger than the smallest constellation, Crux the Southern Cross.

In 1665, Italian astronomer Giovanni Cassini classified Jupiter's Great Red Spot as a per-

manent feature.

The first planetarium opened in Munich in 1923.

In 1916, German physicist Karl Schwarzschild solved Albert Einstein's general theory of relativity field equations and derived the size of the event horizon of a non-rotating black hole.

The International Space Station completes an orbit of Earth in about 90 minutes.

At its equator, Jupiter rotates at 27,900 mph; more than 27 times as fast as the Earth.

Kid Spot Quiz Answers:

1. **Sun, Moon, Earth**
2. **Photosphere**
3. **Syzygy**
4. **Hans Lippershey**

From the Board of Directors

Announcements

- ◇ The 2017 RASC handbooks and calendars are now available!
- ◇ The SJAA bi-annual swap meet event will take place on Sunday, November 13th.
- ◇ The December meeting will take place on 12/3 in lieu of 12/10 due to the YSG Christmas production on 12/10.
- ◇ The club Show-n-Tell event is set for 12/3/2016 after the board meeting.
- ◇ The annual Pot Luck dinner and Awards Ceremony will take place on 2/11/2017 after the board meeting.
- ◇ SJAA Club Officer elections will be held at the 3/11/2017 meeting. There will be two board seats opening up; Ed Wong and Rob Jaworski.
- ◇ The Spring 2017 Swap Meet will be held on Sunday, 3/12/2017

Board Meeting Excerpts

August 20, 2016

In attendance

Dave Ittner, Bill O'Neil, Glenn Newell, Vini Carter, Lee Hoglan, Teruo Utsumi, Ed Wong
Excused: Rob Chapman, Rob Jaworski
Guests: Jose Pereira, Steven Swift, Wolf Witt

Meetup

Dave Ittner provided metrics for Meetup membership (1400) and Active members (440). Discussion about ways to engage the Meetup members ensued.

SJAA/Dr. Robert Armstrong ASP Scholarship

Dave Ittner (on behalf of Rob Jaworski) proposed funding of for the Dr. Armstrong ASP Scholarship Program in the amount of \$700. The motion was approved.

Coyote Valley Harvest & Cambrian Festival Beads

Wolff Witt requested \$200 to purchase beads for kids attending the OSA Coyote Valley Harvest and Cambrian Festival. Motion was made and approved.

September 17, 2016

In attendance

Dave Ittner, Vini Carter, Glenn Newell, Bill O'Neil, Lee Hoglan, Teruo Utsumi, Rob Chapman
Excused: Ed Wong and Rob Jaworski
Guests: George M., Werner K., Muditha K.

Online Member Status Check

Dave Ittner presented on a quick way for Program Leaders to check on member status. Web development is underway by the Coding SIG group.

Member Classification System

Dave Ittner presented the membership classification categories for discussion; New; Qualified; Volunteer. The basic framework was acceptable to the board. Implementation is to be in conjunction with development of the Online Member database by the Coding SIG group.

Loaner Program Expansion

Muditha Kanchana (Kanch) proposed expanding the loaner scope program for the benefit of SJAA members. The presentation available at the SJAA Board Google group:

<https://groups.google.com/d/msg/sjaa-board/2bLKimhl4dw/1a0vBKQNCgAJ>

The general plan includes Phase 1 & 2 implementation over 1-2 years, lays out anticipated costs including new equipment acquisitions, loss and damage. Membership classification would be used to determine level of access to equipment. A Phase 1 budget of \$4,300 was approved for program acquisitions. Additional Phase 2 equipment acquisitions at Kanch's discretion were also approved provided they fall within the Phase 1 budget.

Clean Up Day

Dave Ittner reminded everyone that Sunday, Sept. 25th, 9 am was clean up day. The work will include tossing unusable scopes, S&T Mags, pulling out the 20" scope to prepare for sale and assembling the racks.

October 15, 2016

In attendance

Dave Ittner, Ed Wong, Vini Carter, Glenn Newell, Lee Hoglan, Rob Jaworski, Teruo Utsumi, Rob Chapman
Excused: Bill O'Neil
Guests: Muditha Kanchana, Jeff Baker

SJAA/Dr. Robert Armstrong ASP Scholarship

Rob Jaworski presented a draft of the proposed program for the SJAA / Astronomy Society of the Pacific (ASP) Dr. Robert Armstrong joint scholarship fund. The goal is to support two Southbay teachers per year for their professional development in astronomy. The ASP will select the teacher candidates, provide for development of the teachers and SJAA is to provide funding for the ASP efforts. A 5 year \$3500 budget to fund the program was motioned and approved. Rob will post the final version of the fund upon completion.

Hogue Sound Speaker System

Vini Carter demonstrated an upgraded speaker system for SJAA use at club meeting with guest speakers. The demo was well received so Vini will move forward with pricing of a complete system of speakers, amp and microphones for the club. A DC option for use at Public Star Parties will also be investigated.

Hogue Park Hall Clean Up

Dave Ittner provided an update of the Hogue Hall cleanup and rack effort. It's about 90% complete and looking good, file cabinets and wire racks are in place.

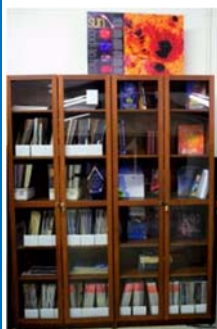
SJAA G4NPs

Dave Ittner updated the group on the Google for non profits (G4NPs) efforts to put in place Shared folders, SJAA contact sheet, Calendars, etc. which are now tied to the SJAA google accounts created for the different programs.

Royal Astronomical Society of Canada (RASC) Guidebook

Teruo Utsumi requested approval for an RASC calendar and handbook purchase based on last years sales. A \$1200 budget for pre-purchase was approved.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change:

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with

only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

(SIG), In February, Bruce Braunstein agreed to lead the SIG group. SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President:	Dave Ittner
Vice President:	Ed Wong
Treasurer:	Rob Jaworski
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Director:	Bill O'Neil
Director:	Teruo Utsumi
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SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

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